

Assignment 8: Building a Family Tree in Prolog

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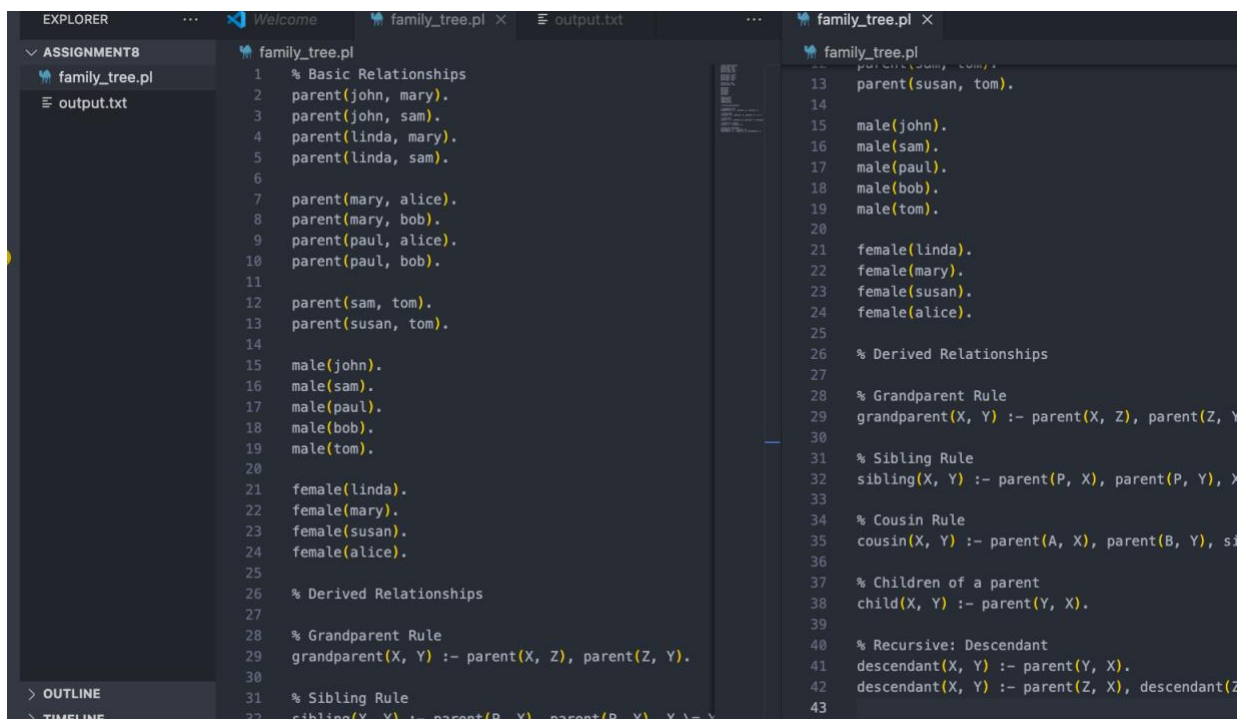
Github: <https://github.com/mbista25742/MSCS-632-M51/tree/main/assignment8>

Abstract:

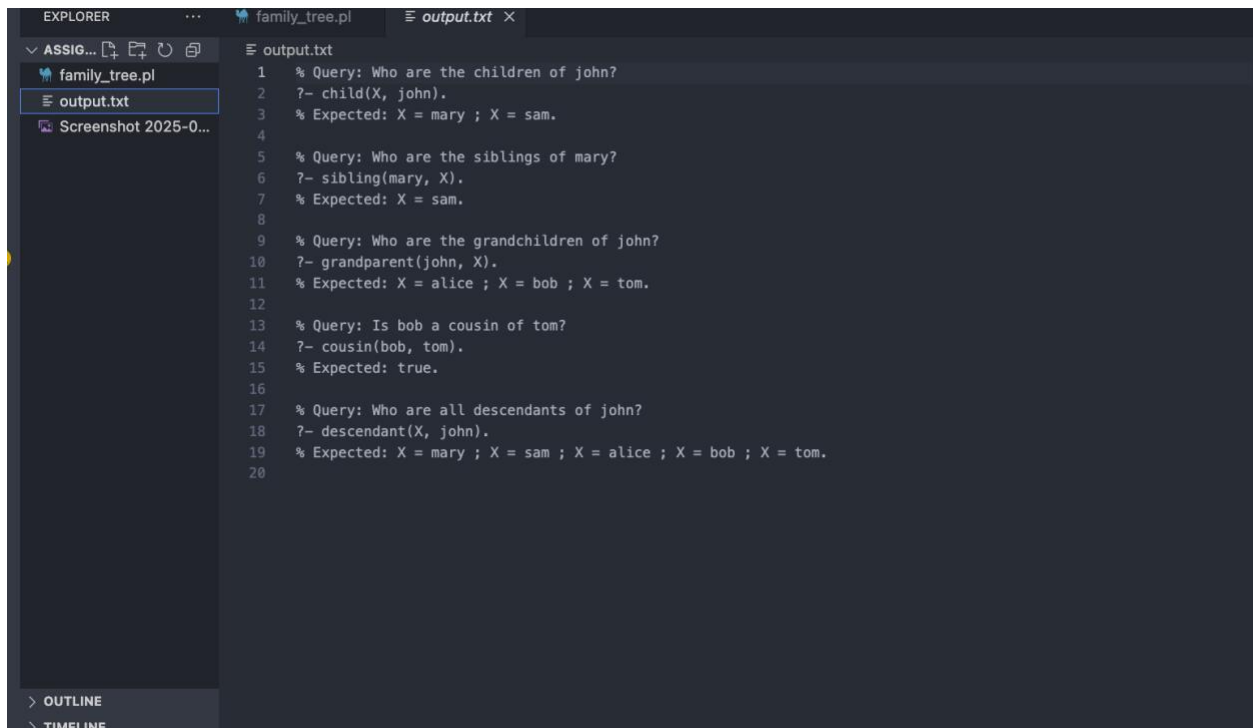
This report documents the creation of a simple family tree program in Prolog. The goal was to define basic and derived familial relationships and enable queries through logical inference and recursion.

Implementation Summary:

Using Prolog, we defined facts for basic relationships like `parent`, `male`, and `female`. Derived relationships such as `sibling`, `grandparent`, and `cousin` were constructed using rule-based logic. Recursive rules, such as `descendant`, demonstrated Prolog's strength in handling indirect relationships.



```
1 % Basic Relationships
2 parent(john, mary).
3 parent(john, sam).
4 parent(linda, mary).
5 parent(linda, sam).
6
7 parent(mary, alice).
8 parent(mary, bob).
9 parent(paul, alice).
10 parent(paul, bob).
11
12 parent(sam, tom).
13 parent(susan, tom).
14
15 male(john).
16 male(sam).
17 male(paul).
18 male(bob).
19 male(tom).
20
21 female(linda).
22 female(mary).
23 female(susan).
24 female(alice).
25
26 % Derived Relationships
27
28 % Grandparent Rule
29 grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
30
31 % Sibling Rule
32 sibling(X, Y) :- parent(P, X), parent(P, Y), X \= Y.
33
34 % Cousin Rule
35 cousin(X, Y) :- parent(A, X), parent(B, Y), A \= B.
36
37 % Children of a parent
38 child(X, Y) :- parent(Y, X).
39
40 % Recursive: Descendant
41 descendant(X, Y) :- parent(Y, X).
42 descendant(X, Y) :- parent(Z, X), descendant(Z, Y).
43
```



The screenshot shows a Prolog IDE with two tabs: `family_tree.pl` and `output.txt`. The `output.txt` tab is active, displaying the results of several Prolog queries. The queries and their outputs are as follows:

```
1 % Query: Who are the children of john?
2 ?- child(X, john).
3 % Expected: X = mary ; X = sam.
4
5 % Query: Who are the siblings of mary?
6 ?- sibling(mary, X).
7 % Expected: X = sam.
8
9 % Query: Who are the grandchildren of john?
10 ?- grandparent(john, X).
11 % Expected: X = alice ; X = bob ; X = tom.
12
13 % Query: Is bob a cousin of tom?
14 ?- cousin(bob, tom).
15 % Expected: true.
16
17 % Query: Who are all descendants of john?
18 ?- descendant(X, john).
19 % Expected: X = mary ; X = sam ; X = alice ; X = bob ; X = tom.
20
```

Challenges Encountered:

The most significant challenge involved structuring the `descendant` rule to avoid infinite recursion. Ensuring uniqueness in relationships like `sibling` and `cousin` also required careful handling of inequalities ($X \neq Y$).

Conclusion:

This exercise highlighted the power of logical inference and recursion in Prolog for modeling complex relationships. The approach offers clarity and flexibility, especially useful in knowledge-based systems.

References:

Sebesta, R. W. (2018). Concepts of Programming Languages (12th ed.). Pearson Education (US). <https://reader2.yuzu.com/books/9780135102251>